

DATA SCIENCE RESOURCES

Financial Projects

Project 1. Loan Approval Prediction

Problem Statement: Banks and NBFCs face high operational costs and default risks due to inaccurate or manual loan approval processes.

Objective: Build a machine learning model to predict whether a loan application should be approved based on applicant profile, income, credit history, and other factors.

Dataset Source

- [Loan Prediction Dataset](#) (Analytics Vidhya)
- [Lending Club open dataset](#)

Techniques & Concepts

- Binary classification
- Feature encoding (categorical → numerical)
- Handling missing values
- Evaluation with ROC-AUC, confusion matrix

Tools & Libraries: Python, Pandas, Scikit-learn, XGBoost, Matplotlib

Expected Outcome: A predictive model that helps automate loan approvals, reducing processing time and improving accuracy.

Project 2. Credit Risk Scoring Model

Problem Statement: Accurately evaluating a customer's creditworthiness is essential to reduce defaults and non-performing assets (NPAs).

Objective: Build a model that scores customer risk into low, medium, or high credit risk tiers using demographic, financial, and behavioral features.

Dataset Source

- [German Credit Risk Dataset \(UCI\)](#)
- [LendingClub loan book](#)

Techniques & Concepts

- Multi-class classification
- Feature importance analysis
- Weight of Evidence (WoE) and Information Value (IV)
- Confusion matrix and ROC analysis

Tools & Libraries: Python, Pandas, Scikit-learn, XGBoost, SHAP, Streamlit (optional for deployment)

Expected Outcome: A credit scoring system that ranks borrowers and informs risk-based interest rates or credit limits.

Project 3. Stock Portfolio Optimization

Problem Statement: Retail and institutional investors often rely on intuition rather than data to build investment portfolios, leading to suboptimal returns.

Objective: Use modern portfolio theory and machine learning to create an optimized stock portfolio with maximum returns for a given level of risk.

Dataset Source

- [Yahoo Finance API via yfinance](#)
- [NSE/BSE stock data](#)

Techniques & Concepts

- Mean-variance optimization (Markowitz Theory)
- Sharpe Ratio and risk-return tradeoff
- Monte Carlo simulation
- Correlation matrix and diversification strategy

Tools & Libraries: Python, NumPy, Pandas, yfinance, Matplotlib, cvxpy or PyPortfolioOpt

Expected Outcome: An optimized portfolio of stocks with calculated weights, expected returns, and risk levels, ideal for algorithmic investing or Robo-advisors.
